

1.9 Idempotent matrices

A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$. Applying it twice has the same effect as applying it once, which is how a projection behaves. Examples are $\mathbf{I}$, $\mathbf{0}$, the centring matrix of Example (Sums of squares as quadratic forms), the products $\mathbf{AG}$ and $\mathbf{GA}$ for a generalized inverse $\mathbf{G}$ (Proposition 1.3.3(a)), and the non-symmetric $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. In linear models, sums of squares are quadratic forms in *symmetric* idempotent matrices, and their degrees of freedom are ranks.

Proposition 1.9.1. General idempotent matrices

Let $\mathbf{A}$ be $n \times n$ and idempotent of rank r .

- (a) Every eigenvalue of $\mathbf{A}$ is 0 or 1.
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$.
- (c) $\mathbf{I} - \mathbf{A}$ is idempotent, $\mathbf{A}(\mathbf{I} - \mathbf{A}) = (\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{0}$, $\mathcal{C}(\mathbf{I} - \mathbf{A}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{I} - \mathbf{A}) = n$.
- (d) An idempotent matrix that is nonsingular equals $\mathbf{I}$.
- (e) $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is idempotent for nonsingular $\mathbf{P}$, and $\mathbf{A}^\top$ is idempotent.

Proof.

- (a) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\lambda\mathbf{x} = \mathbf{A}\mathbf{x} = \mathbf{A}^2\mathbf{x} = \lambda^2\mathbf{x}$, so $\lambda^2 = \lambda$.
- (b) If $r = 0$ both sides vanish. Otherwise take a rank factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ (Theorem 1.2.1). Then $\mathbf{B}\mathbf{F}\mathbf{B}\mathbf{F} = \mathbf{B}\mathbf{F}$. Since $\mathbf{B}$ has full column rank and $\mathbf{F}$ has full row rank, $\mathbf{B}$ has a left inverse and $\mathbf{F}$ a right inverse (Section 1.3), and cancelling them gives $\mathbf{F}\mathbf{B} = \mathbf{I}_r$. So $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{B}\mathbf{F}) = \text{tr}(\mathbf{F}\mathbf{B}) = r$ by Theorem 1.6.1.
- (c) The first two claims are one-line expansions. $\mathcal{C}(\mathbf{I} - \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$ because $\mathbf{A}(\mathbf{I} - \mathbf{A}) = \mathbf{0}$, and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$ gives $\mathbf{v} = (\mathbf{I} - \mathbf{A})\mathbf{v}$. The rank statement is then rank-nullity.
- (d) Multiply $\mathbf{A}^2 = \mathbf{A}$ by $\mathbf{A}^{-1}$.
- (e) $(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^2 = \mathbf{P}^{-1}\mathbf{A}^2\mathbf{P}$ and $(\mathbf{A}^\top)^2 = (\mathbf{A}^2)^\top$.

■

By (c), every $\mathbf{x}$ splits as $\mathbf{A}\mathbf{x} + (\mathbf{I} - \mathbf{A})\mathbf{x}$ with the first part in $\mathcal{C}(\mathbf{A})$ and the second in $\mathcal{N}(\mathbf{A})$. So $\mathbb{R}^n = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ projects onto $\mathcal{C}(\mathbf{A})$ along $\mathcal{N}(\mathbf{A})$. The projection is orthogonal exactly when the two subspaces are orthogonal, which happens iff $\mathbf{A}$ is symmetric (Exercise (B1)).

Theorem 1.9.2. Symmetric idempotent matrices

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix of rank r . Then $\mathbf{A}$ is idempotent iff all its eigenvalues are 0 or 1. In that case

- (a) $\mathbf{A} = \mathbf{Q}_1\mathbf{Q}_1^\top$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns spanning $\mathcal{C}(\mathbf{A})$;
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = r$, the multiplicity of the eigenvalue 1;
- (c) $\mathbf{A}$ and $\mathbf{I} - \mathbf{A}$ are nonnegative definite, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \leq \|\mathbf{x}\|^2$.

Proof. If $\mathbf{A}$ is idempotent its eigenvalues are 0 or 1 by [Proposition 1.9.1\(a\)](#). Conversely, if $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ with each $\lambda_i \in \{0, 1\}$, then $\mathbf{A}^2 = \mathbf{Q}\mathbf{\Lambda}^2\mathbf{Q}^\top = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top = \mathbf{A}$. In that case, order the eigenvalues so that the r ones come first and let $\mathbf{Q}_1$ hold the corresponding eigenvectors. Then $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top = \mathbf{Q}_1\mathbf{Q}_1^\top$, and $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{Q}_1)$ by [Corollary 1.7.3\(a\)](#). This is (a). The trace is the sum of the eigenvalues, which is (b). For (c), the eigenvalues of $\mathbf{A}$ and of $\mathbf{I} - \mathbf{A}$ are nonnegative, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x}$. Also $\|\mathbf{x}\|^2 - \|\mathbf{A}\mathbf{x}\|^2 = \mathbf{x}^\top (\mathbf{I} - \mathbf{A}) \mathbf{x} \geq 0$. ■

[Chapter 6](#) shows that symmetric idempotent matrices are exactly the orthogonal projections, and $\mathbf{Q}_1\mathbf{Q}_1^\top$ in (a) is the projection onto $\mathcal{C}(\mathbf{A})$. In [Chapter 4](#), part (b) becomes the degrees of freedom of a chi-squared distribution.

Theorem 1.9.3. Sums of symmetric idempotent matrices

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (i) $\mathbf{A}$ is idempotent;
- (ii) every $\mathbf{A}_i$ is idempotent;
- (iii) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Any two of these imply the third. Moreover, if $\mathbf{A} = \mathbf{I}_n$ and $\sum_i \text{rank}(\mathbf{A}_i) = n$, then (ii) and (iii) hold.

Proof.

(ii) and (iii) imply (i): $\mathbf{A}^2 = \sum_i \mathbf{A}_i^2 + \sum_{i \neq j} \mathbf{A}_i \mathbf{A}_j = \sum_i \mathbf{A}_i = \mathbf{A}$.

(iii) and (ii) imply (iii): By [Theorem 1.9.2\(c\)](#), $\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x}$ for all $\mathbf{x}$. Fix j and take $\mathbf{x} = \mathbf{A}_j \mathbf{y}$, so that $\mathbf{A}_j \mathbf{x} = \mathbf{x}$ and $\mathbf{x}^\top \mathbf{A}_j \mathbf{x} = \|\mathbf{x}\|^2$. Then

$$\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \mathbf{x}^\top \mathbf{A}_i \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \|\mathbf{A}_i \mathbf{x}\|^2.$$

So $\mathbf{A}_i \mathbf{A}_j \mathbf{y} = \mathbf{A}_i \mathbf{x} = \mathbf{0}$ for every $i \neq j$ and every $\mathbf{y}$.

(iv) and (iii) imply (ii): By (iii), $\mathbf{A} \mathbf{A}_i = \mathbf{A}_i^2$. Using $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{A}_i^2 = \mathbf{A} \mathbf{A}_i = \mathbf{A}^2 \mathbf{A}_i = \mathbf{A} \mathbf{A}_i^2 = \mathbf{A}_i^3$. The eigenvalues of the symmetric $\mathbf{A}_i$ therefore satisfy $\lambda^2 = \lambda^3$, so they lie in $\{0, 1\}$, and $\mathbf{A}_i$ is idempotent by [Theorem 1.9.2](#).

For the last statement, let $r_i = \text{rank}(\mathbf{A}_i)$. For each i , $\mathbf{I} - \mathbf{A}_i = \sum_{j \neq i} \mathbf{A}_j$ has rank at most $\sum_{j \neq i} r_j = n - r_i$ ([Proposition 1.2.3\(d\)](#)). So the eigenvalue 1 of $\mathbf{A}_i$, whose eigenspace is $\mathcal{N}(\mathbf{I} - \mathbf{A}_i)$, has multiplicity at least r_i . But $\mathbf{A}_i$ has exactly r_i nonzero eigenvalues, so all of them equal 1 and $\mathbf{A}_i$ is idempotent. Now (i) holds trivially, and (iii) follows from the second part of the proof. ■

The last statement is the algebraic core of Cochran's theorem ([Chapter 4](#)): if $\mathbf{y}^\top \mathbf{y} = \sum_i \mathbf{y}^\top \mathbf{A}_i \mathbf{y}$ and the ranks add up to n , the quadratic forms are built from mutually orthogonal projections.

1.9.1 Exercises

1.9.1.1 A. Check your understanding

Exercise. A1

Let $\mathbf{A}, \mathbf{B}$ be idempotent of the same size. Show that $\mathbf{AB}$ is idempotent if $\mathbf{AB} = \mathbf{BA}$, and that $\mathbf{A} + \mathbf{B}$ is idempotent iff $\mathbf{AB} + \mathbf{BA} = \mathbf{0}$. Show that the latter forces $\mathbf{AB} = \mathbf{0}$ (symmetry is not needed).

1.9.1.2 B. Practice

Exercise. B1

Let $\mathbf{A}$ be idempotent. Show that $\mathbf{A}$ is symmetric iff $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$.

Solution. Suppose $\mathbf{A}$ is symmetric. For $\mathbf{u} = \mathbf{Ax} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, $\mathbf{u}^\top \mathbf{v} = \mathbf{x}^\top \mathbf{Av} = 0$. Conversely, suppose $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$. For all $\mathbf{x}, \mathbf{y}$, $(\mathbf{I} - \mathbf{A})\mathbf{y} \in \mathcal{N}(\mathbf{A})$, so $\mathbf{x}^\top \mathbf{A}^\top (\mathbf{I} - \mathbf{A})\mathbf{y} = 0$. Hence $\mathbf{A}^\top (\mathbf{I} - \mathbf{A}) = \mathbf{0}$, that is, $\mathbf{A}^\top = \mathbf{A}^\top \mathbf{A}$. The right side is symmetric, so $\mathbf{A}^\top$ is symmetric, and so is $\mathbf{A}$.

1.9.1.3 C. Going deeper

Exercise. C1

Let $\mathbf{A}$ and $\mathbf{B}$ be symmetric idempotent of the same order. Show that the following are equivalent: (i) $\mathbf{A} - \mathbf{B}$ is nonnegative definite; (ii) $\mathbf{A} - \mathbf{B}$ is idempotent; (iii) $\mathbf{AB} = \mathbf{B}$; (iv) $\mathbf{BA} = \mathbf{B}$; (v) $\mathcal{C}(\mathbf{B}) \subseteq \mathcal{C}(\mathbf{A})$. Deduce that a difference of two sums of squares is again a sum of squares exactly when one model is nested in the other, the situation of [Chapter 11](#).

Solution. First note that $\mathbf{Ax} = \mathbf{x}$ for every $\mathbf{x} \in \mathcal{C}(\mathbf{A})$: if $\mathbf{x} = \mathbf{Ay}$, then $\mathbf{Ax} = \mathbf{A}^2\mathbf{y} = \mathbf{Ay} = \mathbf{x}$.

(v) $\Rightarrow$ (iii): each column of $\mathbf{B}$ lies in $\mathcal{C}(\mathbf{B}) \subseteq \mathcal{C}(\mathbf{A})$ and is therefore fixed by $\mathbf{A}$, so $\mathbf{AB} = \mathbf{B}$.

(iii) $\Rightarrow$ (v): $\mathcal{C}(\mathbf{B}) = \mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$. (iii) $\Leftrightarrow$ (iv): transpose, using the symmetry of $\mathbf{A}$ and $\mathbf{B}$.

(iii) $\Rightarrow$ (ii): with $\mathbf{AB} = \mathbf{BA} = \mathbf{B}$, $(\mathbf{A} - \mathbf{B})^2 = \mathbf{A} - \mathbf{AB} - \mathbf{BA} + \mathbf{B} = \mathbf{A} - \mathbf{B}$. (ii) $\Rightarrow$ (i): $\mathbf{A} - \mathbf{B}$ is symmetric and idempotent, hence nonnegative definite by [Theorem 1.9.2\(c\)](#).

(i) $\Rightarrow$ (iii): let $\mathbf{x} \in \mathcal{C}(\mathbf{B})$, so $\mathbf{Bx} = \mathbf{x}$ and $\mathbf{x}^\top \mathbf{Bx} = \|\mathbf{x}\|^2$. Then $\mathbf{x}^\top \mathbf{Ax} \geq \mathbf{x}^\top \mathbf{Bx} = \|\mathbf{x}\|^2$, while $\mathbf{x}^\top \mathbf{Ax} \leq \|\mathbf{x}\|^2$ by [Theorem 1.9.2\(c\)](#). So $\mathbf{x}^\top (\mathbf{I} - \mathbf{A})\mathbf{x} = 0$, and since $\mathbf{I} - \mathbf{A}$ is nonnegative definite, Proposition 1.7.5(b) gives $(\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$, that is, $\mathbf{Ax} = \mathbf{x}$. Applying this to the columns of $\mathbf{B}$ gives $\mathbf{AB} = \mathbf{B}$.

For the last statement, if $\mathbf{A}$ and $\mathbf{B}$ are the projections attached to two models, then $\mathbf{y}^\top \mathbf{Ay} - \mathbf{y}^\top \mathbf{By}$ is again a quadratic form in a symmetric idempotent matrix — a sum of squares with its own degrees of freedom $\text{rank}(\mathbf{A}) - \text{rank}(\mathbf{B})$ — exactly when $\mathcal{C}(\mathbf{B}) \subseteq \mathcal{C}(\mathbf{A})$.

Exercise. C2

Let $\mathbf{A}$ and $\mathbf{B}$ be symmetric idempotent with $\mathbf{AB} = \mathbf{BA}$. Show that $\mathbf{AB}$ is symmetric idempotent with $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A}) \cap \mathcal{C}(\mathbf{B})$, and that $\mathbf{A} + \mathbf{B} - \mathbf{AB}$ is symmetric idempotent with column space $\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$; deduce $\text{rank}(\mathbf{A} + \mathbf{B} - \mathbf{AB}) = \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - \text{rank}(\mathbf{AB})$. Give two symmetric idempotent matrices of order 2 that do not commute, and show that for them $\mathbf{AB}$ is not a projection onto the intersection.

Solution. Symmetry: $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top = \mathbf{BA} = \mathbf{AB}$. Idempotence: $(\mathbf{AB})^2 = \mathbf{A}(\mathbf{BA})\mathbf{B} = \mathbf{A}^2\mathbf{B}^2 = \mathbf{AB}$. If $\mathbf{x} \in \mathcal{C}(\mathbf{AB})$ then $\mathbf{x} = \mathbf{A}(\mathbf{B}\mathbf{y}) \in \mathcal{C}(\mathbf{A})$ and, using $\mathbf{AB} = \mathbf{BA}$, also $\mathbf{x} = \mathbf{B}(\mathbf{A}\mathbf{y}) \in \mathcal{C}(\mathbf{B})$. Conversely, if $\mathbf{x}$ lies in both column spaces, then $\mathbf{A}\mathbf{x} = \mathbf{x}$ and $\mathbf{B}\mathbf{x} = \mathbf{x}$, so $\mathbf{AB}\mathbf{x} = \mathbf{x}$ and $\mathbf{x} \in \mathcal{C}(\mathbf{AB})$.

Put $\mathbf{N} = \mathbf{A} + \mathbf{B} - \mathbf{AB}$. It is symmetric, and expanding with $\mathbf{AB} = \mathbf{BA}$ and $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{B}^2 = \mathbf{B}$ gives $\mathbf{N}^2 = \mathbf{A} + \mathbf{AB} - \mathbf{AB} + \mathbf{AB} + \mathbf{B} - \mathbf{AB} - \mathbf{AB} - \mathbf{AB} + \mathbf{AB} = \mathbf{A} + \mathbf{B} - \mathbf{AB}$. Its column space lies in $\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$; conversely $\mathbf{N}\mathbf{x} = \mathbf{x}$ for $\mathbf{x} \in \mathcal{C}(\mathbf{A})$ (then $\mathbf{N}\mathbf{x} = \mathbf{x} + \mathbf{B}\mathbf{x} - \mathbf{B}\mathbf{x}$) and for $\mathbf{x} \in \mathcal{C}(\mathbf{B})$, so $\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B}) \subseteq \mathcal{C}(\mathbf{N})$. The rank statement is [Proposition 1.9.1\(b\)](#) applied to the three idempotent matrices, since the trace is linear.

For a counterexample take $\mathbf{A} = \mathbf{e}_1\mathbf{e}_1^\top$ and $\mathbf{B} = \frac{1}{2}\mathbf{J}_2$, the projections onto two different lines. Then $\mathbf{AB} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{BA} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ differ, $\mathbf{AB}$ is neither symmetric nor idempotent, and $\mathcal{C}(\mathbf{A}) \cap \mathcal{C}(\mathbf{B}) = \{\mathbf{0}\}$ although $\mathbf{AB} \neq \mathbf{0}$.

Exercise. C3

Let $\mathbf{A}$ be idempotent of order n and rank r , not necessarily symmetric. Show that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ for some nonsingular $\mathbf{P}$, and conversely. Read off from this a second proof that $\text{tr}(\mathbf{A}) = r$, and find the characteristic polynomial of $\mathbf{A}$ and the dimensions of its eigenspaces.

Solution. By [Proposition 1.9.1\(c\)](#), $\mathbb{R}^n = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A})$, with the two dimensions r and $n - r$. Take a basis $\mathbf{p}_1, \dots, \mathbf{p}_r$ of $\mathcal{C}(\mathbf{A})$ and a basis $\mathbf{p}_{r+1}, \dots, \mathbf{p}_n$ of $\mathcal{N}(\mathbf{A})$; together they are a basis of $\mathbb{R}^n$, so $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$ is nonsingular. Every $\mathbf{x} \in \mathcal{C}(\mathbf{A})$ satisfies $\mathbf{A}\mathbf{x} = \mathbf{x}$, as in [Exercise \(C1\)](#), so $\mathbf{A}\mathbf{p}_i = \mathbf{p}_i$ for $i \leq r$ and $\mathbf{A}\mathbf{p}_j = \mathbf{0}$ for $j > r$. Column by column this says $\mathbf{AP} = \mathbf{P}\begin{pmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$. Conversely, a matrix similar to that block matrix is idempotent by [Proposition 1.9.1\(e\)](#), and has the same rank.

Since the trace is unchanged by similarity (Eq. 1.6.1), $\text{tr}(\mathbf{A}) = r$, which proves [Proposition 1.9.1\(b\)](#) again without a rank factorization. Similar matrices have the same characteristic polynomial ([Proposition 1.7.1\(c\)](#)), so $\det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^{n-r}(\lambda - 1)^r$. The eigenspaces are $\mathcal{C}(\mathbf{A})$ for the eigenvalue 1 and $\mathcal{N}(\mathbf{A})$ for the eigenvalue 0, of dimensions r and $n - r$: an idempotent matrix is diagonalizable even when it is not symmetric.